441/1
AGRICULTURE PAPER 1
MARKING SCHEME

SECTION A

1. **Physical agents of weathering**
   - Wind
   - Temperature
   - Moving water/rainfall
   - Moving ice/glacier **½ x3 =1 ½ mks**

2. **Methods of breaking seed dormancy**
   - Hot water treatment
   - Scarification/filing/nicking
   - Partial burning
   - Soaking in water
   - Chemical method/use of concentrated sulphuric acid/ potassium nitrate. **½ x3 =1 ½ mks**

3. **Signs of nematode attack on crops**
   - Stunted growth
   - Wilting
   - Discolouration of foliage
   - Gal formation/root knots **½ x3 =1 ½ mks**

4. **Net Revenue** is the difference between total revenue and total costs i.e. profit while **marginal Revenue** is extra revenue earned by selling one more unit of output. **1x1=1 mk**

5. **Uses of the following in compost manure**
   Top soil- introduces organisms that help on decomposition.
   Wood ash – increases level of potassium phosphorus
   -Neutralises acids produced due to decomposition
   Organic manure – Provides food and shelter to micro-organisms **½ x3 =1 ½ mks**

6. **Advantages of using polythene sleeves to raise seedlings**
   - Prevents roots disturbance during transplanting
   - Easy to transport and handle prevents attack by soil-borne pests and diseases
   - Transplanting can be delayed awaiting condusive conditions **½ x3 =1 ½ mks**

7. **Disadvantages of communal land temperature system**
   - Yields are low due to overstocking
   - Difficult to control parasites and diseases
   - Inbreeding as a result of uncontrolled breeding
   - Long term planning and investment cannot be done
   - There is easy spread of parasites /pests and diseases **½ x3 =1 ½ mks**
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8. **Methods of improving productivity of farm labour**
   - Training of workers
   - Mechanizing farm operations
   - Providing incentives to workers
   - Supervision and counseling of workers
   - Reducing length of working period
   - Promoting individual qualities of workers
   - Assigning specific work task **½ x4 = 2mks**

9. **Factors determining choice of an irrigation system**
   - Topography
   - Type of soil
   - Type of crop grown
   - Funds available/costs involved/capital
   - Size of land to be irrigated
   - Profitability/viability of irrigation
   - Availability of water/source of water/quality of water/rainfall pattern **½ x4 =2mks**

10. **Characteristics of shifting cultivation**
   - When soil fertility goes down, crops are not grown until fertility is restored
   - Plenty of arable land is available
   - Practicable with annual crops
   - Crops are grown for subsistence
   - Farming systems are rarely used
   - Simple farming techniques are used **½ x4 =2mks**

11. **Thinning** is the removal of excess seedlings from a seedbed while **pricking out** is the removal of excess seedlings from a nursery bed then transferring them to another nursery bed.
    1 x1=1mk

12. **Books of account**
   - Cash book
   - Ledger
   - Journal
   - Inventory **½ x3 =1 ½ mks**

13. **Cultural methods of weed control**
   - Early planting
   - Mulching
   - Deep ploughing
   - Crop rotation
   - Use of clean implements
   - Use of cover crops
   - Use of clean planting materials **½ x3 =1 ½ mks**

14. **Precautions observed when harvesting pyrethrum**
   - Do not compact flowers
   - Put flowers in wooden baskets
   - Do not pick wet flowers
   - Pick flowers with 2-3 rows of disc florets fully open
   - Deliver flowers to factory within 24hrs of harvesting **½ x3 =1 ½ mks**

15. **Advantages of minimum tillage**
   - Conserves soil moisture
   - Cheaper
   - Maintains soil fertility
   - Saves labour
   - Saves time
   - Maintains soil structure
   - Reduces soil erosion
   - Preserves useful soil organisms **½ x4 =1 2mks**

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16. **Diseases caused by viruses to crop**
    - Maize streak
    - Cassava mosaic
    - Tobacco mosaic
    - Ratoon stunting
    - Citrus tristeza/Greening diseases **½ x3 =1 ½ mks**

17. Examples of joint products in crop production
- Wheat and straw
- Sugar and molasses
- Pyrethrum and pyrethrum marc
- Cotton limit and cotton seeds
- Copra and coir **½ x3 =1 ½ mks**

18. **Advantages of land fragmentation**
    - Allows diversification of production
    - Spreads production throughout
    - Eases crop rotation
    - Reduces disputes over traditional land **½ x3 =1 ½ mks**

19. **Advantages of zero grazing**
    - Requires small piece of land
    - Reduces wastage of feeds
    - Higher production
    - Reduces destruction of properties by livestock
    - Quicker accumulation of manure
    - Easy control of parasites and diseases
    - Allows higher stocking rate. **½ x4 =1 ½ mks**

SECTION B

20. **(a) 1** – Top soil
3 – Weathered rock
4 – Parent rock/Bedrock **½ x3 =1 ½ mks**

(b) **B** – Prismatic structure
C – Collumnar structure **½ x2 =1mk**

(c) soil erosion/denudation **½ x1= ½ mk**

(d) – Impedes drainage
- Reduces aeration
- Lowers root development **½ x2 =1mk**

(e) - Overcultivation
- Burning of land
- Monocropping
- Working soil at wrong moisture content
- Repeated use of heavy machinery **½ x2 =1mk**

21. (a) **D** – Whip/Tongue grafting
E – Aerial laying/macrootting **½ x2 =1mk**

(b) **5** – Scion
6 – Root stock **½ x2 =1mk**

(c) When a large number of propagules are required
When the stem cannot easily be bent **1 x1 =1mk**

(d) Characteristics of a good root stock
- Healthy
- Resistant/Tolerant to soil-borne pests and diseases
- Adaptable to different soil conditions e.g. PH
- Compatible with different scions **½ x4 =1 ½ mks**

22. (a) **Method of drainage**
- Cambered bed/Raised bed/Ridging **1 x1 =1mk**

(b) **Other methods of drainage**

- Pumping water
- Planting appropriate trees/trees that use up a lot of water
- Open ditches
- Underground drain pipes
- French drains **1 x2 =2mks**

(c) **Reasons for drainage**

- Raises soil temperature
- Increases soil volume
- Prevents soil erosion
- Improves aeration
- Improves microbial activities
- Reduces salt accumulation
- Controls vectors **1 x2 =2mks**

23. (a) 100 kg CAN = 20kg of N
200kg CAN = (x) kg of N √

$$\Rightarrow x = \frac{20 \times 200}{100}$$

x = 40kg N√

OR

$$\frac{x \times 100}{200} = 20\%$$

$$\frac{100x}{200} = 20$$

$$100x = 200 \times 20$$ **1 x3 =3mks**

$$x = \frac{4000}{100}$$

$$x = 40kgN$$

(b) Methods of top-dressing

- Broadcasting
- Bund application
- Spot application
- Ring application

SECTION C

24. (a) **Problems facing agriculture in Kenya**

- Lack of capital for investment
- Pests and diseases
- Shortage of skilled personel
- Shortage of labour
- Shortage of extension services
- Shortage of land
- Poor attitudes towards physical work in the farm
- Drastic climatic changes
- Natural calamities
- Lack of markets
- Delayed payments
- Shortage of farming inputs **1x10=10mks**

(b) (i) Importance of soil air

- Improves microbial activities
- Provides oxygen necessary for plant root respiration
- Provides nitrogen required by nitrogen fixing organisms **1x3=3mks**

(ii) Importance of mineral mater

- Provides Nutrients
- Provides anchorage to crops **1x2=2mks**

(c) uses of water in the farm

- Cooling of engines
- Irrigation/watering of plants
- Fish rearing
- Mixing chemicals
- Mixing concrete
- Domestic use
- Watering livestock **1x5mks**

25. (a) **Factors leading to loss of soil fertility**

(i) Soil erosion
- Loss of nutrients
- Shallow soil depth

(ii) Leaching
- Loss of soil nutrients

(iii) Monocropping
- Building of pests and diseases
- Over-utilization of certain nutrients

(iv) Continuous cropping
- Continuous removal of nutrients from the fields

(v) Change of soil PH
- Reduces availability of certain nutrients

(vi) Burning
- Increases volatilization
- Destroys soil nutrients/structure

(vii) Salt accumulation
- Changes soil PH
- Removal of water from plant

(viii) Denitrification
- Loss of nitrogen from the soil

(ix) Excessive infestation by soil borne pests, diseases and weeds
- Destroy crops
- Deprive crops of nutrients

Statement of factor 1x5 = 5mks
Explanation of factor 1x5 = 5mks

(b) **Factors affecting efficiency of herbicides**

i. Stage of growth of weeds
- Younger weeds are more killed

ii. Mode of action
- Systemic are more effective

iii. Concentration of herbicides
- The more concentrated the more effective

iv. Formulation of herbicides
- Oil formulation if more effective

v. Method of application
- Selective application increases efficiency

vi. Leaf angle
- Horizontal leaves are more killed

vii. Nature of leaf surface
- Hairy leaves holds more herbicides and thus more killed

viii. Location of growth points
- Hidden growth points are less affected

ix. Rooting system
- Aerial roots are better killed

x. Presence of specialized structures
- Weeds with underground permeating organs are less affected

xi. Size of spray drops
- Smaller droplets are more retained by the weeds making them more effective

xii. Climatic conditions
- Wind and rain reduce efficiency of herbicides.

Statement of factor 1x5 = 5mks
Explanation of factor 1x5 = 5mks

26. (a) **Roles of co-operative societies**

- Sharing overhead costs
- Advancing credit to members
- Bargain for better prices
- Provide easy access to inputs
- Eliminate exploitation of members by middlemen
- Members get services of trained and experienced staff
- Offers technical advice to members
- Transports and sell farmers' produce
- Offer improved storage
- Offer improved processing
- Collects and assemble produce
- Offer farm machinery to members on hire terms
- Share profits to members as dividends/bonuses
- Grading of produce

1x10=10mks

(b) **How farmers adjusts to risks and uncertainties**

- Diversification production
- Selecting more certain enterprises
- Input rationing
- Insuring production
- Production on contrast
- Flexibility in production methods
- Using modern production methods
- Building sound networth
- Employing security personnel
- Joining co-operative unions and farmers unions
- Keeping to proven production techniques

1 x 10 = 10mks